

Muhammad Zaid Mohsin

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EDUCATION

Ghulam Ishaq Khan Institute - Bachelor of Science in Computer Science

September 2022 – October 2026

CGPA: 3.66 GIK Merit Scholarship Winner, **3-time** Dean's Honar Award Winner

Relevant Coursework

Introduction to Artificial Intelligence, Cloud Computing, Database Management System, Computer Communication and Networks, Operating Systems, Data Structures and Algorithms, Software Engineering.

PROFESSIONAL AND LEADERSHIP EXPERIENCE

Backend Developer, Talpur Foundation Pakistan

January 2025 – February 2025

- Designed and developed scalable **Django-based backend architecture**, implementing RESTful APIs and database optimizations for enhanced performance.
- Developed and optimized Django-based APIs, improving backend **response time by 40%** and reducing **database queries by 30%**.
- Orchestrated the deployment and hosting on **Vercel**, delivering a highly reliable and scalable platform.
- Launched the live website at <https://talpurfoundationpakistan.com/>, exemplifying modern web development practices.

Lead Organizer, AWS Pakistan Session– GIKI

September 2024– Present

- Led a collaboration between AWS Pakistan and GIKI, organizing a workshop on **AI in web development and AWS services**, strengthening industry-academia ties and established relations for future collaboration with AWS Pakistan for the first time.
- Coordinated an engaging session for **40+ students** at GIKI focused on practical applications of AWS services, receiving positive feedback from over **90% of participants** regarding content relevance to current industry practices.
- Successfully managed** event logistics, including venue coordination, participant registration, and session execution, contributing to a **highly rated and well-received event**.

Machine Learning Internee, Digital Empowerment Network

July 2024 – August 2024

- Completed a 4-week internship at Digital Empowerment Network (DEP) as a Machine Learning Intern, gaining hands-on experience in practical machine learning applications

Society of Mechanical Engineering Pakistan, GIKI

January 2023 – Present

- Organized **3 HEC-certified workshops** in Swabi universities, attended by **150+ students**.

PROJECTS

AI Powered Customer Relationship Model (CRM) Database Management System

Python, Streamlit, OpenAI, PostgreSQL

- Advanced a comprehensive Client Management System (CMS) using Django and PostgreSQL, integrating project, employee, contact, meeting, and leads management into a single platform, **reducing administrative overhead by an estimated 40%** for business owners through streamlined workflows.
- Implemented an NLP-driven chatbot within the CMS, capable of **resolving 75% of user queries** related to project updates, employee details, and client information in real time, **decreasing manual support efforts by 50%** and enhancing user productivity.

AI-Based Course Recommendation System Artificial Intelligence

Python, Streamlit, OpenAI

- Proceeded with a personalized course recommendation system utilizing Random Forest and content-based filtering, achieving an **RMSE of 0.258 and MSE of 0.066** in predicting user ratings for Coursera courses, delivering highly accurate and tailored learning suggestions.
- Engineered a scalable framework to process and analyze a simulated dataset of **1,000 users** alongside Coursera course data, showcasing the system's capability to handle diverse user profiles and its potential for expansion to resources like videos, articles, and books.
- Enhanced the online learning experience by providing precise course recommendations based on user preferences and educational goals, potentially cutting course selection time by up to **50%**, **improving efficiency** and relevance for learners.

Deepfake Video Detection Tool Using Neural Networks Computer Vision

Python, Django, TensorFlow, OpenCV

- Initiated a deepfake detection system using **ResNext CNN for feature extraction and LSTM RNN** for sequence classification.
- Trained model on **6000+ videos (FaceForensics++, Celeb-DF)**, achieving **97.76% accuracy on FaceForensics++ with 100-frame sequences**.
- Architected a scalable video processing model that efficiently analyzes sequences ranging from 10 to 100 frames, **delivering 89.35% accuracy** on a mixed dataset with 40 frames, optimized for real-time performance across various video types and resolutions.

TECHNICAL SKILLS

Languages: C++, C, HTML, Python, SQL, Verilog, Assembly Language, Bash.

Tools and Technologies: AWS, Azure, Vercel, Linux (Ubuntu), Windows, Git, Jupyter Notebook, PostgreSQL, LaTeX/Overleaf,

Soft Skills & Leadership: Public Speaking, Teamwork, Leadership, Content Writing.

Libraries: pandas, NumPy, Matplotlib, tensorflow, OpenCV.

Frameworks & Tools: Django, Flask

AI & Machine Learning: NLP, APIs, LLMs, Supervised Learning, Unsupervised Learning, Model Deployment,